**Concept Question 10-1:** What are the advantages and disadvantages of elliptical satellite orbits in comparison to the geostationary orbit?

**Concept Question 10-2:** Why do satellite communication systems use different frequencies for the uplink and downlink segments? Which segment uses the higher frequency and why?

**Concept Question 10-3:** How does the use of antenna polarization increase the number of channels carried by the communication system?

**Concept Question 10-4:** What are the sources of noise that contribute to the total system noise temperature of a receiver?

## 10-5 Radar Sensors

The term ***radar*** is a contracted form of the phrase ***radio detection and ranging***, which conveys some—not all—of the features of a modern radar system. Historically, radar systems were first developed and used at *radio* frequencies, including the microwave band, but we now also have light radars or ***lidars*** that operate at optical wavelengths. Over the years, the name radar has lost its original meaning and has come to signify any *active* electromagnetic sensor that uses its own source to illuminate a region of space and then measure the echoes generated by reflecting objects contained in that region. In addition to detecting the presence of a reflecting object and determining its *range* by measuring the time delay of short-duration pulses transmitted by the radar, a radar is also capable of specifying the *position* of the target and its radial velocity. Measurement of the ***radial velocity*** of a moving object is realized by measuring the ***Doppler frequency shift*** produced by the object. Also, the strength and shape of the reflected pulse carry information about the shape and material properties of the reflecting object.

Radar is used for a wide range of civilian and military applications, including air traffic control, aircraft navigation, law enforcement, control and guidance of weapon systems, remote sensing of Earth's environment, weather observation, astronomy, and collision avoidance for automobiles. The frequency bands used for the various types of radar applications extend from the megahertz region to frequencies as high as 225 GHz.

### 10-5.1 Basic Operation of a Radar System

The block diagram shown in **Fig. 10-9** contains the basic functional elements of a pulse radar system. The ***synchronizer–modulator*** unit serves to synchronize the operation of the transmitter and the ***videoprocessor–display unit***; it does so by generating a train of direct-current (dc) narrow-duration, evenly spaced pulses. These pulses, which are supplied to both the transmitter, and the videoprocessor–display unit, specify the times at which radar pulses are transmitted. The transmitter contains a high-power radio-frequency (RF) oscillator with an on/off control voltage actuated by the pulses supplied by the synchronizer–modulator unit. Hence, the transmitter generates pulses of RF energy equal in duration and spacing to the dc pulses generated by the synchronizer–modulator unit. Each pulse is supplied to the antenna through a ***duplexer***, which

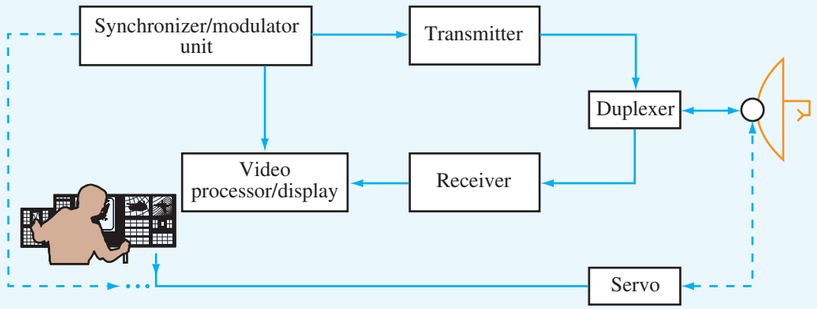



**Figure 10-9** Basic block diagram of a radar system.





**Concept Question 10-1:** What are the advantages and disadvantages of elliptical satellite orbits in comparison to the geostationary orbit?

**Concept Question 10-2:** Why do satellite communication systems use different frequencies for the uplink and downlink segments? Which segment uses the higher frequency and why?

**Concept Question 10-3:** How does the use of antenna polarization increase the number of channels carried by the communication system?

**Concept Question 10-4:** What are the sources of noise that contribute to the total system noise temperature of a receiver?

## 10-5 雷达传感器

雷达一词是无线电探测和测距一词的缩写形式，它传达了现代雷达系统的部分（而非全部）特征。从历史上看，雷达系统首先是在射频（包括微波频段）下开发和使用的，但我们现在也有在光波长下运行的光雷达或激光雷达。多年来，雷达这个名字已经失去了它原来的含义，并开始表示任何有源电磁传感器，这些传感器使用自己的源来照亮空间区域，然后测量该区域内反射物体所产生的回波。除了检测反射物体的存在之外通过测量雷达发射的短时脉冲的时间延迟来确定其范围，雷达还能够指定目标的位置及其径向速度。运动物体径向速度的测量是通过测量物体产生的多普勒频移来实现的。此外，反射脉冲的强度和形状还携带有关反射物体的形状和材料特性的信息。

雷达具有广泛的民用和军事应用，包括空中交通管制、飞机导航、执法、武器系统的控制和制导、地球环境遥感、天气观测、天文学和汽车防撞。用于各种类型雷达应用的频段从兆赫兹区域延伸到高达 225 GHz 的频率。

### 10-5.1 雷达系统的基本操作

图 10-9 所示的框图包含脉冲雷达系统的基本功能元件。同步器-调制器单元用于同步发射机和视频处理器-显示单元的操作；它通过生成一系列窄持续时间、均匀间隔的直流 (dc) 脉冲来实现这一点。这些脉冲提供给发射器和视频处理器-显示单元，指定雷达脉冲的发射时间。发射器包含一个高功率射频 (RF) 振荡器，具有由同步器-调制器单元提供的脉冲驱动的开/关控制电压。因此，发射器生成的射频能量脉冲的持续时间和间隔与同步器-调制器单元生成的直流脉冲相同。每个脉冲都通过双工器提供给天线，

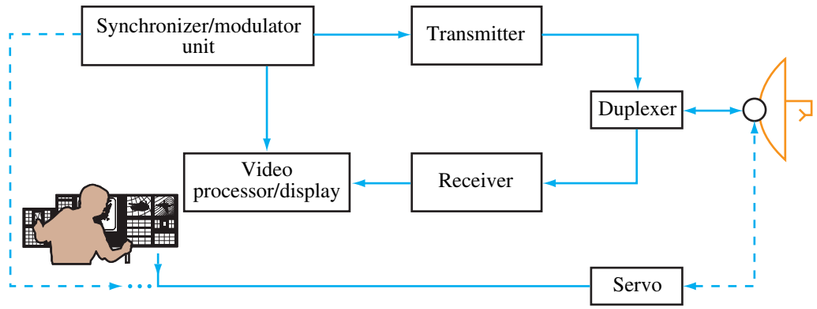



图 10-9 雷达系统的基本框图。

allows the antenna to be shared between the transmitter and the receiver. The duplexer, which often is called the ***transmitter/receiver*** (/R) ***switch***, first connects the transmitter to the antenna for the duration of the pulse and then connects the antenna to the receiver for the remaining period until the start of a new pulse. Some duplexers, however, are passive devices that perform the sharing and isolation functions continuously. The circulator shown in **Fig. 10-5** is an example of a passive duplexer. After transmission by the antenna, a portion of the transmitted signal is intercepted by a reflecting object (often called a ***target***) and scattered in many directions. The energy reradiated by the target back toward the radar is collected by the antenna and delivered to the receiver, which processes the signal to detect the presence of the target and to extract information on its location and velocity. The receiver converts the reflected RF signals into lower-frequency video signals and supplies them to the videoprocessor–display unit, which displays the extracted information in a format suitable for the intended application. The servo unit positions the orientation of the antenna beam in response to control signals provided by either an operator, a control unit with preset functions, or a control unit commanded by another system. The control unit of an air-traffic-control radar, for example, commands the servo to rotate the antenna in azimuth continuously. In contrast, the radar antenna placed in the nose of an aircraft is made to scan back and forth over only a specified angular sector.

## 10-5.2 Unambiguous Range

The collective features of the energy transmitted by a radar are called the ***signal waveform***. For a pulse radar, these features include (1) the carrier frequency $f$, (2) the ***pulse length*** $\tau$, (3) the ***pulse repetition frequency*** $f_p$ (number of pulses per second) or equivalently the ***interpulse period*** $T_p = 1/f_p$, and (4) the modulation (if any) within the pulses. Three of these features are illustrated in **Fig. 10-10**. Modulation, which refers to control of the amplitude, frequency, or phase of the signal, is beyond the level of the present treatment.

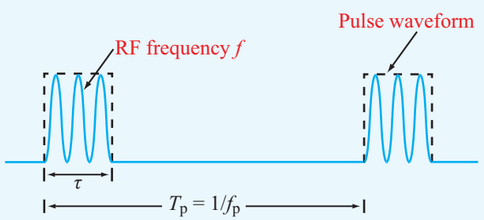



**Figure 10-10** A pulse radar transmits a continuous train of RF pulses at a pulse repetition frequency $f_p$.

The range of a target is determined by measuring the time delay $T$ taken by the pulse to travel to the target and back. For a target at range $R$,

$$T = \frac{2R}{c}, \tag{10.13}$$

where $c = 3 \times 10^8$ m/s is the speed of light and the factor 2 accounts for the two-way propagation. The maximum target range that a radar can measure unambiguously, called the ***unambiguous range*** $R_u$, is determined by the interpulse period $T_p$ and is given by

$$R_u = \frac{cT_p}{2} = \frac{c}{2f_p}. \tag{10.14}$$

The range $R_u$ corresponds to the maximum range that a target can have such that its echo is received before the transmission of the next pulse. If $T_p$ is too short, an echo signal due to a given pulse might arrive after the transmission of the next pulse, in which case the target would appear to be at a much shorter range than it actually is.

According to Eq. (10.14), if a radar is to be used to detect targets that are as far away as 100 km, for example, then $f_p$ should be less than 1.5 kHz, and the higher the pulse repetition frequency (PRF), the shorter is the unambiguous range $R_u$. Consideration of $R_u$ alone suggests selecting a low PRF, but other considerations suggest selecting a very high PRF. As we will see later in Section 10-6, the signal-to-noise ratio of the radar receiver is directly proportional to $f_p$. Hence, it would be advantageous to select a PRF as high as possible. Moreover, in addition to determining the maximum unambiguous range $R_u$, the PRF also determines the maximum Doppler frequency (hence, the target's maximum radial velocity) that the radar can measure unambiguously. If the requirements on maximum range and velocity cannot be met by the same PRF, then some compromise may be necessary. Alternatively, it is possible to use a ***multiple-PRF*** radar system that transmits a few pulses at one PRF followed by another series of pulses at another PRF. Then the two sets of received pulses are processed together to remove the ambiguities that would have been present with either PRF alone.

## 10-5.3 Range and Angular Resolutions

Consider a radar observing two targets located at ranges $R_1$ and $R_2$, as shown in **Fig. 10-11**. Let $t = 0$ denote the time corresponding to the start of the transmitted pulse. The pulse length is $\tau$. The return due to target 1 will arrive at $T_1 = 2R_1/c$ and will have a length $\tau$ (assuming that the pulse length in





允许发射器和接收器之间共享天线。双工器通常称为发射器/接收器 (/R) 开关，首先在脉冲持续时间内将发射器连接到天线，然后在剩余时间内将天线连接到接收器，直到新脉冲开始。然而，一些双工器是连续执行共享和隔离功能的无源设备。图10-5所示的循环器是无源双工器的示例。通过天线传输后，部分传输信号被反射物体（通常称为目标）拦截并向多个方向散射。目标向雷达再辐射的能量由天线收集并传送到接收器，接收器处理信号以检测目标的存在并提取有关其位置和速度的信息。接收器将反射的射频信号转换为低频视频信号，并将其提供给视频处理器-显示单元，该单元以适合预期应用的格式显示提取的信息。伺服单元响应于操作员、具有预设功能的控制单元或由另一系统命令的控制单元提供的控制信号来定位天线波束的方向。例如，空中交通管制雷达的控制单元命令伺服系统连续旋转天线的方位角。相比之下，放置在飞机机头的雷达天线只能在指定的角度区域内来回扫描。

## 10-5.2 明确范围

雷达发射的能量的集体特征称为信号波形。对于脉冲雷达，这些特征包括 (1) 载波频率 f 、(2) 脉冲长度 τ、(3) 脉冲重复频率 $f_p$（每秒脉冲数）或等效的脉冲间周期 $T_p = 1/f_p$，以及 (4) 脉冲内的调制（如果有）。图 10-10 展示了其中三个特征。调制，指的是

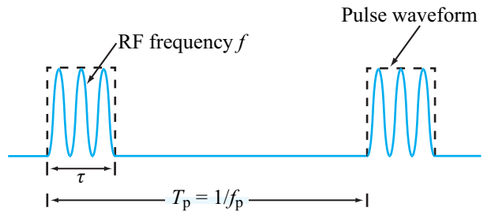



图 10-10 脉冲雷达以脉冲重复频率 $f_p$ 发射连续的射频脉冲。

控制信号的幅度、频率或相位超出了现有治疗的水平。

目标的范围是通过测量脉冲传播到目标并返回的时间延迟 T 来确定的。对于 R 范围内的目标，

$$T = \frac{2R}{c}, \tag{10.13}$$

其中 $c = 3 \times 10^8$ m/s 是光速，因子 2 说明双向传播。雷达可以明确测量的最大目标距离，称为明确距离 $R_u$，由脉冲间周期 $T_p$ 确定，并由下式给出

$$R_u = \frac{cT_p}{2} = \frac{c}{2f_p}. \tag{10.14}$$

范围$R_u$对应于目标可以具有的最大范围，以便在发送下一个脉冲之前接收到其回波。如果 $T_p$ 太短，则给定脉冲产生的回波信号可能会在下一个脉冲传输后到达，在这种情况下，目标看起来会比实际距离短得多。

根据方程。根据式（10.14），如果雷达要用于探测远至100公里的目标，则$f_p$应小于1.5kHz，并且脉冲重复频率（PRF）越高，明确距离$R_u$越短。单独考虑 $R_u$ 建议选择较低的 PRF，但其他考虑因素建议选择非常高的 PRF。正如我们稍后将在第 10-6 节中看到的，雷达接收器的信噪比与 $f_p$ 成正比。因此，选择尽可能高的 PRF 是有利的。此外，除了确定最大明确范围 $R_u$ 之外，PRF 还确定雷达可以明确测量的最大多普勒频率（因此，目标的最大径向速度）。如果相同的 PRF 无法满足最大范围和速度的要求，则可能需要进行一些折衷。或者，可以使用多 PRF 雷达系统，该系统在一个 PRF 处发射几个脉冲，然后在另一个 PRF 处发射另一系列脉冲。然后，对两组接收到的脉冲进行一起处理，以消除单独使用任一 PRF 时可能出现的模糊性。

## 10-5.3 距离和角度分辨率

考虑雷达观测位于 $R_1$ 和 $R_2$ 范围内的两个目标，如图 10-11 所示。令t = 0 表示对应于发射脉冲开始的时间。脉冲长度为τ。目标 1 的返回将到达 $T_1 = 2R_1/c$，并且将具有长度 τ（，假设脉冲长度为

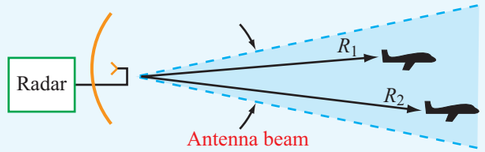



**Figure 10-11** Radar beam viewing two targets at ranges $R_1$ and $R_2$.

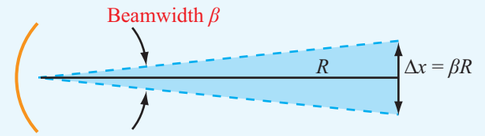



**Figure 10-12** The azimuth resolution $\Delta x$ at a range $R$ is equal to $\beta R$.

space is much greater than the radial extent of the target). Similarly, the return due to target 2 will arrive at $T_2 = 2R_2/c$. The two targets are resolvable as distinct targets so long as $T_2 \geq T_1 + \tau$ or, equivalently,

$$\frac{2R_2}{c} \geq \frac{2R_1}{c} + \tau. \tag{10.15}$$

The ***range resolution*** of the radar, $\Delta R$, is defined as the minimum spacing between two targets necessary to avoid overlap between the echoes from the two targets. From Eq. (10.15), this occurs when

$$\Delta R = R_2 - R_1 = c\tau/2. \tag{10.16}$$

Some radars are capable of transmitting pulses as short as 1 ns in duration or even shorter. For $\tau = 1$ ns, $\Delta R = 15$ cm.

The basic angular resolution of a radar system is determined by its antenna beamwidth $\beta$, as shown in **Fig. 10-12**. The corresponding ***azimuth resolution*** $\Delta x$ at a range $R$ is given by

$$\Delta x = \beta R, \tag{10.17}$$

where $\beta$ is in radians. In some cases, special techniques are used to improve the angular resolution down to a fraction of the beamwidth. One example is the monopulse tracking radar described in Section 10-8.

## 10-6 Target Detection

Target detection by radar is governed by two factors: (1) the ***signal energy*** received by the radar receiver due to reflection of part of the transmitted energy by the target and (2) the ***noise energy*** generated by the receiver. **Figure 10-13** depicts the output response of a radar receiver as a function of time and shows the signals due to two targets displayed against the noise contributed by external sources as well as by the devices making up the receiver. The random variations exhibited by the noise may at times make it difficult to distinguish the signal reflected by the target from a noise spike. In **Fig. 10-13**, the mean noise-power level at the receiver output is denoted by $P_{no} = G_{rec}P_{ni}$, where $G_{rec}$ is the receiver gain and $P_{ni}$ is the noise level referred to the receiver's input terminals. The power levels $P_{r_1}$ and $P_{r_2}$ represent the echoes of the two targets observed by the radar. Because of the random nature of noise, it is necessary to set a threshold level, $P_{r_{min}}$, for detection. For threshold level 1 indicated in **Fig. 10-13**, the radar will produce the presence of both targets, but it will also detect a ***false alarm***. The chance of this occurring is called the ***false-alarm probability***. On the other hand, if the threshold level is raised to level 2 to avoid the false alarm, the radar will not detect the presence of the first target. A radar's ability to detect the presence of a target is characterized by a ***detection probability***. The setting of the threshold signal level relative to the mean noise level is thus made on the basis of a compromise that weighs both probabilities.

To keep the noise level at a minimum, the receiver is designed so that its bandwidth $B$ is barely wide enough to pass most of the energy contained in the received pulse. Such a design, called a ***matched filter***, requires that $B$ be equal to the reciprocal of the pulse length $\tau$ (i.e., $B = 1/\tau$). Hence, for a matched-filter receiver, Eq. (10.10) becomes

$$P_{ni} = KT_{sys}B = \frac{KT_{sys}}{\tau}. \tag{10.18}$$

The signal power received by the radar, $P_r$, is related to the transmitted power level, $P_t$, through the radar equation. We will first derive the radar equation for the general case of a ***bistatic radar*** configuration in which the transmitter and receiver are not necessarily at the same location, and then we will specialize the results to the ***monostatic radar*** case where the transmitter and receiver are colocated. In **Fig. 10-14**, the target is at range $R_t$ from the transmitter and at range $R_r$ from the receiver. The power density illuminating the target is given by

$$S_t = \frac{P_t}{4\pi R_t^2} G_t \qquad (\mathrm{W/m^2}), \tag{10.19}$$

where $(P_t/4\pi R_t^2)$ represents the power density that would have been radiated by an isotropic radiator, and $G_t$ is the gain of the





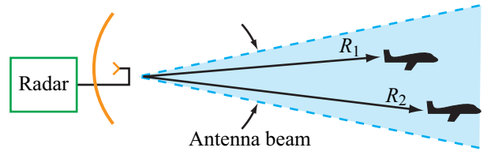



图 10-11 雷达波束观察距离 $R_1$ 和 $R_2$ 处的两个目标。

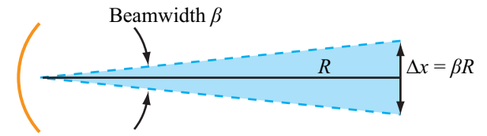



图 10-12 距离 R 处的方位角分辨率 Δx 等于 β R。

空间远大于目标的径向范围）。类似地，目标 2 的返回将到达 $T_2 = 2R_2/c$。只要 $T_2 \geq T_1 + \tau$ 或等价地，这两个目标就可以解析为不同的目标

$$\frac{2R_2}{c} \geq \frac{2R_1}{c} + \tau. \tag{10.15}$$

雷达的距离分辨率 ΔR 被定义为避免两个目标的回波重叠所需的两个目标之间的最小间距。从方程。 （10.15），这发生在

$$\Delta R = R_2 - R_1 = c\tau/2. \tag{10.16}$$

有些雷达能够发射持续时间短至 1 ns 甚至更短的脉冲。对于 $\tau = 1$ ns，$\Delta R = 15$ cm。

雷达系统的基本角分辨率由其天线波束宽度β 决定，如图10-12所示。范围 R 处相应的方位角分辨率 Δx 由下式给出

$$\Delta x = \beta R, \tag{10.17}$$

其中 β 的单位是弧度。在某些情况下，可以使用特殊技术将角分辨率提高到波束宽度的一小部分。一个例子是第 10-8 节中描述的单脉冲跟踪雷达。

## 10-6 目标检测

雷达的目标检测受两个因素控制：(1) 雷达接收器由于目标反射部分发射能量而接收到的信号能量；(2) 接收器产生的噪声能量。图 10-13 描述了雷达接收器随时间变化的输出响应，并显示了两个目标产生的信号，以及外部源以及构成接收器的设备产生的噪声。噪声表现出的随机变化有时可能会导致难以区分目标反射的信号和噪声尖峰。在图10-13中，接收机输出端的平均噪声功率电平用$P_{no} = G_{rec}P_{ni}$表示，其中$G_{rec}$是接收机增益，$P_{ni}$是接收机输入端子的噪声电平。功率水平 $P_{r_1}$ 和 $P_{r_2}$ 代表雷达观测到的两个目标的回波。由于噪声的随机性，需要设置一个阈值水平$P_{r_{min}}$来进行检测。对于图 10-13 所示的阈值级别 1，雷达将产生两个目标的存在，但它也会检测到误报。发生这种情况的几率称为误报概率。另一方面，如果将阈值级别提高到级别 2 以避免误报，雷达将不会检测到第一个目标的存在。雷达检测目标存在的能力由检测概率来表征。因此，相对于平均噪声水平的阈值信号水平的设置是基于权衡两种概率的折衷而进行的。

为了将噪声水平保持在最低水平，接收器的设计使其带宽 B 的宽度仅足以通过接收到的脉冲中包含的大部分能量。这种设计称为匹配滤波器，要求 B 等于脉冲长度 τ ( 的倒数，即 $B = 1/\tau$)。因此，对于匹配滤波器接收器，方程： (10.10) 变为

$$P_{ni} = KT_{sys}B = \frac{KT_{sys}}{\tau}. \tag{10.18}$$

雷达接收到的信号功率 $P_r$ 通过雷达方程与发射功率电平 $P_t$ 相关。我们将首先针对双基地雷达配置的一般情况推导雷达方程，其中发射机和接收机不一定位于同一位置，然后我们将结果专门针对发射机和接收机位于同一位置的单基地雷达情况。在图10-14中，目标距离发射机的距离为$R_t$，距离接收机的距离为$R_r$。照射目标的功率密度由下式给出

$$S_t = \frac{P_t}{4\pi R_t^2} G_t \qquad (\mathrm{W/m^2}), \tag{10.19}$$

其中 $(P_t/4\pi R_t^2)$ 表示各向同性辐射器辐射的功率密度，$G_t$ 是各向同性辐射器的增益

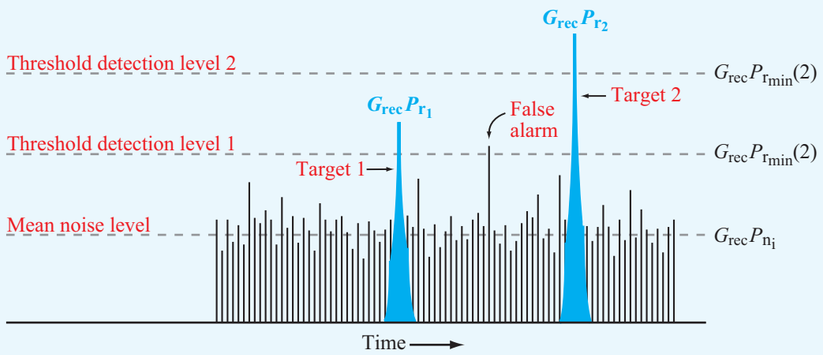



**Figure 10-13** The output of a radar receiver as a function of time.

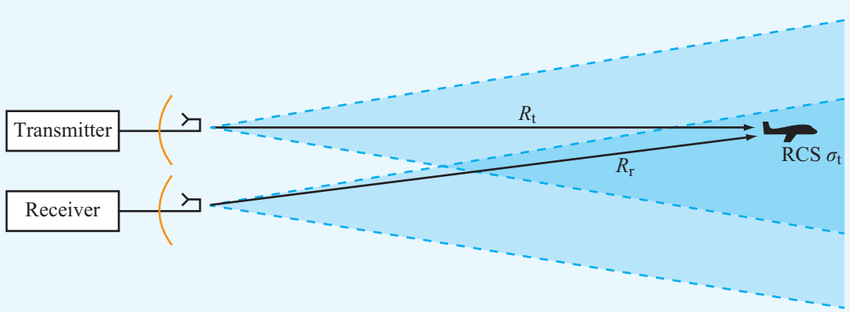



**Figure 10-14** Bistatic radar system viewing a target with radar cross section (RCS) $\sigma_t$.

transmitting antenna in the direction of the target. The target is characterized by a ***radar cross section*** (***RCS***) $\sigma_t$ ($m^2$), defined so that the power intercepted and then reradiated by the target is

$$P_{rer} = S_t\sigma_t = \frac{P_tG_t\sigma_t}{4\pi R_t^2} \quad \text{(W)}. \tag{10.20}$$

This reradiated power spreads out over a spherical surface, resulting in a power density $S_r$ incident upon the receiving radar antenna. Hence,

$$S_r = \frac{P_{rer}}{4\pi R_r^2} = \frac{P_tG_t\sigma_t}{(4\pi R_tR_r)^2} \quad (W/m^2). \tag{10.21}$$

With an effective area $A_r$ and radiation efficiency $\xi_r$, the receiving radar antenna intercepts and delivers (to the receiver) power $P_r$ given by

$$P_r = \xi_rA_rS_r = \frac{P_tG_t\xi_rA_r\sigma_t}{(4\pi R_tR_r)^2} = \frac{P_tG_tG_r\lambda^2\sigma_t}{(4\pi)^3R_t^2R_r^2}, \tag{10.22}$$

where we have used Eqs. (9.29) and (9.64) to relate the effective area of the receiving antenna, $A_r$, to its gain $G_r$. For a monostatic antenna that uses the same antenna for the transmit and receive functions, $G_t = G_r = G$ and $R_t = R_r = R$. Hence,

$$P_r = \frac{P_tG^2\lambda^2\sigma_t}{(4\pi)^3R^4}. \quad \textbf{(radar equation)} \tag{10.23}$$

Unlike the one-way communication system for which the dependence on $R$ is as $1/R^2$, the range dependence given by



444 CHAPTER 10 SATELLITE COMMUNICATION SYSTEMS AND RADAR SENSORS

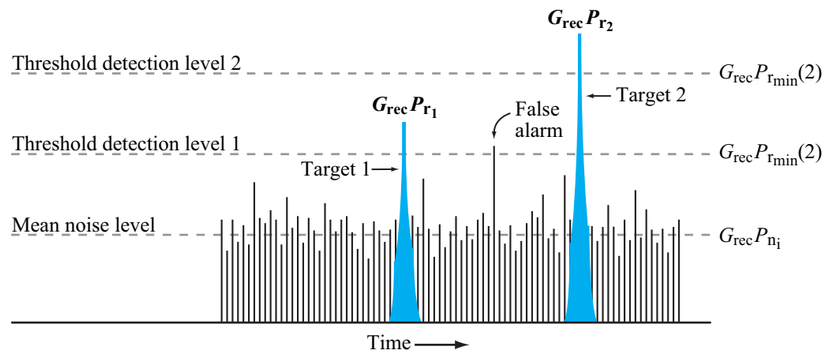



图 10-13 雷达接收器的输出与时间的函数关系。

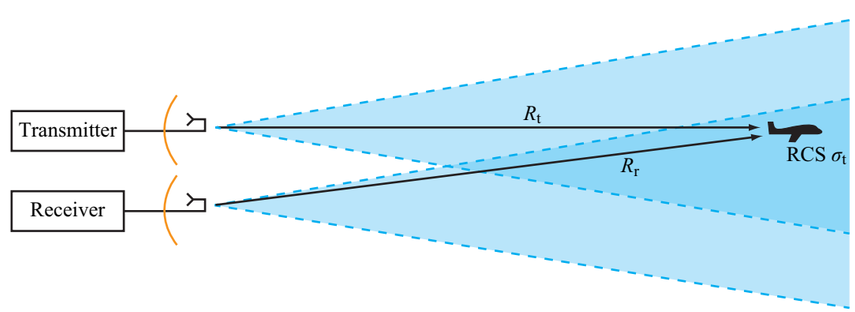



图10 -14 双基地雷达系统用雷达截面观察目标（R CS) $\sigma_t$。

发射天线指向目标方向。目标由雷达横截面 (RCS) $\sigma_t$ ($m^2$) 来表征，定义为目标拦截然后再辐射的功率为

$$P_{rer} = S_t\sigma_t = \frac{P_tG_t\sigma_t}{4\pi R_t^2} \quad \text{(W)}。 \tag{10.20}$$

这种再辐射功率在球面上扩散，导致接收雷达天线上的功率密度 $S_r$。因此，

$$S_r = \frac{P_{rer}}{4\pi R_r^2} = \frac{P_tG_t\sigma_t}{(4\pi R_tR_r)^2} \quad (W/m^2). \tag{10.21}$$

凭借有效面积 $A_r$ 和辐射效率 $\xi_r$，接收雷达天线拦截并传送（至接收器）

功率 $P_r$ 由下式给出

$$P_r = \xi_rA_rS_r = \frac{P_tG_t\xi_rA_r\sigma_t}{(4\pi R_tR_r)^2} = \frac{P_tG_tG_r\lambda^2\sigma_t}{(4\pi)^3R_t^2R_r^2}, \tag{10.22}$$

我们使用等式的地方。（9.29）和（9.64）将接收天线的有效面积 $A_r$ 与其增益 $G_r$ 联系起来。对于使用同一天线实现发射和接收功能的单基地天线，$G_t = G_r = G$ 和 $R_t = R_r = R$。因此，

$$P_r = \frac{P_tG^2\lambda^2\sigma_t}{(4\pi)^3R^4}. \quad \textbf{(radar equation)} \tag{10.23}$$

与对 R 的依赖性为 $1/R^2$ 的单向通信系统不同，范围依赖性由下式给出：

the ***radar equation*** goes as $1/R^4$, which is the product of two one-way propagation processes.

The detection process may be based on the echo from a single pulse or on the addition (integration) of echoes from several pulses. We will consider only the single-pulse case here. A target is said to be detectable if its echo signal power $P_r$ exceeds $P_{r_{min}}$, which is the ***threshold detection level*** indicated in **Fig. 10-13**. The ***maximum detectable range*** $R_{max}$ is the range beyond which the target cannot be detected, corresponding to the range at which $P_r = P_{r_{min}}$ in Eq. (10.23). Thus,

$$R_{max} = \left[\frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 P_{r_{min}}}\right]^{1/4}. \tag{10.24}$$

The signal-to-noise ratio is equal to the ratio of the received signal power $P_r$ to the mean input noise power $P_{ni}$ given by Eq. (10.18):

$$S_n = \frac{P_r}{P_{ni}} = \frac{P_r \tau}{KT_{sys}}, \tag{10.25}$$

and the ***minimum signal-to-noise ratio*** $S_{min}$ corresponds to when $P_r = P_{r_{min}}$:

$$S_{min} = \frac{P_{r_{min}} \tau}{KT_{sys}}. \tag{10.26}$$

Use of Eq. (10.26) in Eq. (10.24) gives

$$R_{max} = \left[\frac{P_t \tau G^2 \lambda^2 \sigma_t}{(4\pi)^3 KT_{sys} S_{min}}\right]^{1/4}. \tag{10.27}$$

The product $P_t\tau$ is equal to the energy of the transmitted pulse. Hence, according to Eq. (10.27), *it is the energy of the transmitted pulse rather than the transmitter power level alone that determines the maximum detectable range.* A high-power narrow pulse and an equal-energy, low-power long pulse will yield the same radar performance as far as maximum detectable range is concerned. However, the range-resolution capability of the long pulse is much poorer than that of the short pulse [see Eq. (10.16)].

The maximum detectable range $R_{max}$ also can be increased by improving the signal-to-noise ratio. This can be accomplished by integrating the echoes from multiple pulses in order to increase the total amount of energy received from the target. The number of pulses available for integration over a specified integration time is proportional to the PRF. Hence, from the standpoint of maximizing target detection, it is advantageous to use as high a PRF as allowed by other considerations.

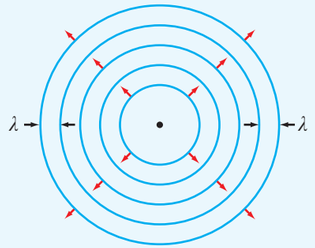
## 10-7 Doppler Radar

The Doppler effect is a shift in the frequency of a wave caused by the motion of the transmitting source, the reflecting object, or the receiving system. As illustrated in **Fig. 10-15**, a wave radiated by a stationary isotropic point source forms equally spaced concentric circles as a function of time travel from the source. In contrast, a wave radiated by a moving source is compressed in the direction of motion and is spread out in the opposite direction. Compressing a wave shortens its wavelength, which is equivalent to increasing its frequency. Conversely, spreading it out decreases its frequency. The change in frequency is called the ***Doppler frequency shift*** $f_d$. That is, if $f_t$ is the frequency of the wave radiated by the moving source, the frequency $f_r$ of the wave that would be observed by a stationary receiver is

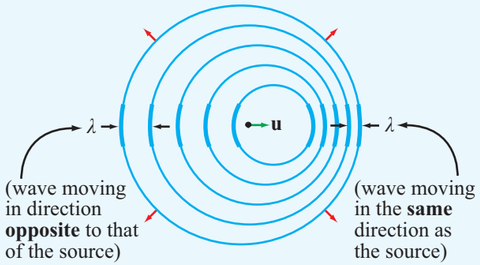
$$f_r = f_t + f_d. \tag{10.28}$$

The magnitude and sign of $f_d$ depend on the direction of the velocity vector relative to the direction of the range vector connecting the source to the receiver.

Consider a source transmitting an electromagnetic wave




(a) Stationary source




(b) Moving source

**Figure 10-15** A wave radiated from a point source when (a) stationary and (b) moving. The wave is compressed in the direction of motion, spread out in the opposite direction, and unaffected in the direction normal to motion.





雷达方程为 $1/R^4$，它是两个单向传播过程的乘积。

检测过程可以基于来自单个脉冲的回波或基于来自多个脉冲的回波的相加（积分）。我们在这里仅考虑单脉冲情况。如果目标的回波信号功率 $P_r$ 超过 $P_{r_{min}}$（图 10-13 所示的阈值检测水平），则称目标是可检测的。最大可检测范围$R_{max}$是超过该范围就无法检测到目标的范围，对应于式(1)中$P_r = P_{r_{min}}$的范围。（10.23）。因此，

$$R_{max} = \left[\frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 P_{r_{min}}}\right]^{1/4}. \tag{10.24}$$

信噪比等于接收信号功率 $P_r$ 与等式 1 给出的平均输入噪声功率 $P_{ni}$ 之比。（10.18）：

$$S_n = \frac{P_r}{P_{ni}} = \frac{P_r \tau}{KT_{sys}}, \tag{10.25}$$

而最小信噪比$S_{min}$对应于当$P_r = P_{r_{min}}$时：

$$S_{min} = \frac{P_{r_{min}} \tau}{KT_{sys}}. \tag{10.26}$$

使用等式。(10.26) 式(10.24) 给出

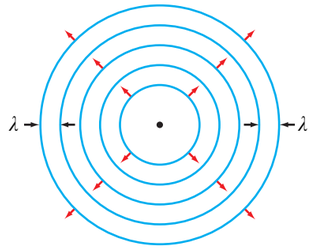
$$R_{max} = \left[\frac{P_t \tau G^2 \lambda^2 \sigma_t}{(4\pi)^3 KT_{sys} S_{min}}\right]^{1/4}. \tag{10.27}$$

乘积 $P_t\tau$ 等于发射脉冲的能量。因此，根据等式。根据方程（10.27），确定最大可检测范围的是发射脉冲的能量，而不是发射器的功率电平。就最大可探测范围而言，高功率窄脉冲和等能量、低功率长脉冲将产生相同的雷达性能。然而，长脉冲的距离分辨率能力比短脉冲差得多[见方程1]。(10.16)]。

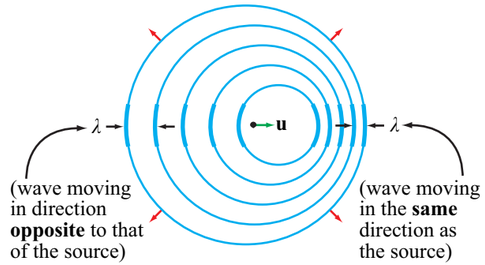
最大可检测范围$R_{max}$也可以通过提高信噪比来增加。这可以通过对多个脉冲的回波进行积分来实现，以增加从目标接收到的能量总量。在指定的积分时间内可用于积分的脉冲数量与 PRF 成正比。因此，从最大化目标检测的角度来看，使用其他考虑因素允许的尽可能高的 PRF 是有利的。

## 10-7 多普勒雷达

多普勒效应是由发射源、反射物体的运动引起的波频率变化，




(a) Stationary source




(b) Moving source

图 10-15 当 (a) 静止和 (b) 移动时从点源辐射的波。波在运动方向上被压缩，在相反方向上展开，并且在垂直于运动的方向上不受影响。

或接收系统。如图 10-15 所示，由静止各向同性点源辐射的波形成等距同心圆，作为从源传播的时间的函数。相反，移动源辐射的波在运动方向上被压缩，并在相反方向上扩散。压缩波会缩短其波长，相当于增加其频率。相反，将其展开会降低其频率。频率的变化称为多普勒频移 $f_d$。也就是说，如果 $f_t$ 是移动源辐射的波的频率，则静止接收器观测到的波的频率 $f_r$ 为

$$f_r = f_t + f_d. \tag{10.28}$$

$f_d$ 的大小和符号取决于速度矢量相对于连接源和接收器的距离矢量方向的方向。

考虑发射电磁波的源